

College of Industrial Technology
King Mongkut's University of Technology North Bangkok

Seat No.

Midterm Examination of Semester 2

Year: 2019

Subject: 341131 Electrical Drawing I

Section: 5-6

Date: 21 January 2020

Time: 15.00-17.00

Name: _____ ID: _____ Class: _____

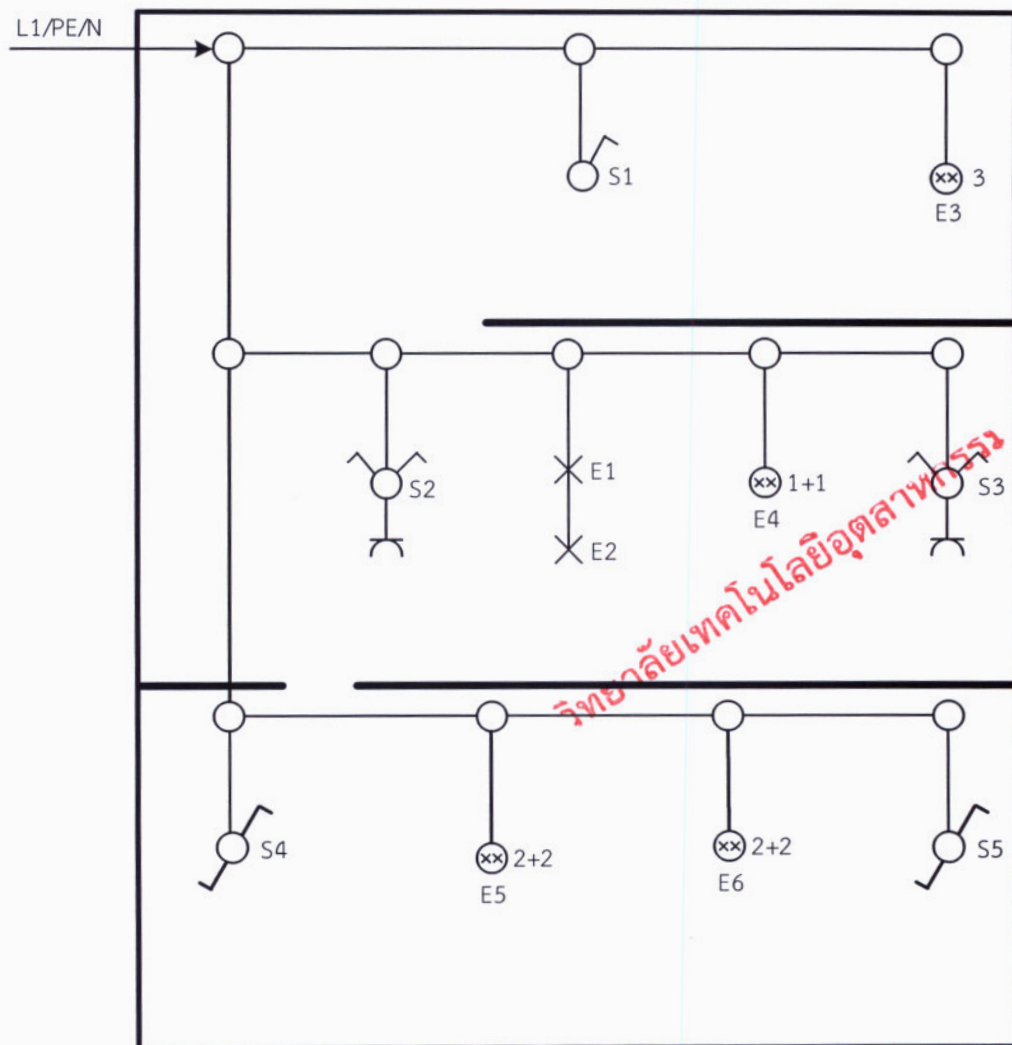
Directions: The test is designed to measure your comprehension. There will be 5 pages (including this page) and they are worth 40 points.

- Answer the questions on the answer sheet.
- Books, documents and lecture notes are not allowed.
- You must be in the room for one hour after the exam is started and, while taking the exam, you cannot go out except in an emergency case.
- Before leaving, make sure you do not bring this test outside.
- Do not use any electronic communication device.
- Calculators can NOT be used in this test.

Now begin the test.

Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University.

1. The floor plan with electrical circuit diagrams is shown below. The diagram includes 2 incandescent lamps (E1 and E2), 4 series incandescent lamps (E3, E4, E5 and E6), 1 switch (S1), 2 series switches (S2 and S3), 2 two-way switches (S4 and S5) and 2 plug sockets. Please draw the wiring, one-line and control diagrams. (40 points).



Remarks:

- E3 is controlled by S1.
- E1 and E2 are controlled by S2.
- E4 is controlled by S3.
- E5 and E6 that are connected in parallel are controlled by S4 and S5.
- 2 plug sockets are independent of each other.

Answer Sheet

Name: _____ ID: _____ Class: _____

One-line diagram (10 points)

เลขที่นั่งสอบ

วิทยาลัยเทคโนโลยีอุตสาหกรรม

Wiring diagram (20 points) Name..... ID.....



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Name: _____ ID: _____ Class: _____

Control diagram (10 points)

วิทยาลัยเทคโนโลยีอุตสาหกรรม

Assoc.Prof.Dr.Rattanakorn Phadungthin

Asst.Prof.Dr.Pinanta Chatwattana